Question 1.

a) 5 certificates, 3 folders, $n=5, k=3$.

$\lceil \frac{n}{k} \rceil = \lceil \frac{5}{3} \rceil$

$= 2$.

$\therefore$ At least 2 certificates must be in the same folder.

b) Type of pets : cat, dog, goat $\rightarrow$ 3

$\boxed{\text{WORST CASE :}}$ $\rightarrow \lceil \frac{4}{3} \rceil$

1st person : cat
2nd person : dog
3rd person : goat.

For the first 3 people can have a different pet. The next people (4th person) must have the same type of pet that match one of them. The minimum number is 4.

Question 2.

a) Total 4-digit codes using digits 0–9 ; $10 \times 10 \times 10 \times 10 = 10000$

To get exactly 2 distinct digits ; ${}^{10}C_2 = 45$.

$2 \times 2 \times 2 \times 2 = 16$ codes.

$16 - 2 = 14$ valid codes.

Probability $= \frac{14 \times 45}{10000} = \frac{630}{10000} = 0.063$.

b) i. $P(X \geq 2) = P(X=2) + P(X=3) = 0.4 + 0.2$
$= 0.6$.

ii. $P(3 > X \geq 0) = P(X=0) + P(X=1) + P(X=2)$
$= 0.1 + 0.3 + 0.4$
$= 0.8$.

Question 3.

a) $P(D) = 0.05$ $\quad P(N) = 1 - 0.05 = 0.95$ $\quad P(R|D) = 0.9$ $\quad P(R|N) = 0.1$

$P(R) = P(R|D)P(D) + P(R|N)P(N)$
$= 0.9(0.05) + 0.1(0.95)$
$= 0.14$.

b) $P(D|R) = \frac{P(R|D)P(D)}{P(R)}$

$= \frac{0.9(0.05)}{0.14}$

$= 0.3214$

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Question 4.

a) $P(E) = \frac{3}{4} = 0.75$

b) $P(F) = \frac{2}{4} = \frac{1}{2}$
$= 0.5.$

c) $P(E \wedge F) = \frac{2}{4} = \frac{1}{2}$
$= 0.5$

d) $P(E \wedge F) = \frac{1}{2} = 0.5$

$P(E) \times P(F) = 0.75 \times 0.5$
$= 0.375$

$P(E \wedge F) \neq P(E) \times P(F)$

$\therefore$ E and F are not independent.

Question 5

a) $P(D|A) = 0.02$
$P(D|B) = 0.04$
$P(D|C) = 0.01.$
$P(A) = 0.4$
$P(B) = 0.35$
$P(C) = 0.25$

$P(D) = P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C)$
$= 0.02(0.4) + 0.04(0.35) + 0.01(0.25)$
$= 0.0245.$

b) $P(B|D) = \frac{P(D|B)P(B)}{P(D)} = \frac{0.04(0.35)}{0.0245.}$
$= 0.5714.$

Question 6.

a) $P(S) = 0.15$
$P(L) = 1 - 0.15$
$= 0.85$
$P(F|S) = 0.95$
$P(F|L) = 0.08$

$P(F) = P(F|S)P(S) + P(F|L)P(L)$
$= 0.95(0.15) + 0.08(0.85)$
$= 0.2105.$

b) $P(S|F) = \frac{P(F|S)P(S)}{P(F)} = \frac{0.95(0.15)}{0.2105}$
$= 0.6770.$

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